

PIPE NAME : 4-SPW-9405003-S98-15N_P04A-01

PIPE MODIFIED : 15.44.14 JAN 2025, Drawing Created at 15:46:48 15 January 2025

135



Project PROVIDENCE
Approval For Construction
PIPING MANAGER : Bea Chang Won
2026-01-21

LOTTE BIOLOGICS



PROJECT PROVIDENCE

LINE NO.	PROJ. NO.	AREA	PDB	CWA NO.	PUAN
4-SPW-9405003-S98-15N	16480	FLUID			SPW

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PPE ENVA ASTM A1521S16L SCH15S BE RT-100%	1.31	598	S98	P	PA02-0000184
2	PPE ENVA ASTM A1521S16L SCH15S BE RT-100%	2.12	0.41	S98	P	PA02-0000182
3	HAIF COUPLING SW CL3000 ASTM A182 F316S16L	4.31	1	S98	TH	PR09-0000351
4	PPE NPPLE PRE SMLD ASTM A1521S16L SCH15S 10MM(4.7)	3/4"	1	S98	NPPG	PR09-0000458
5	TEE BWN WELDED ASTM A1521S16L SCH15S	4.31	1	S98	TE	PR09-0000352
6	ECC REDUCER BWN WELDED ASTM A1521S16L SCH15S	4.31	1	S98	RE	PR09-0000352
7	90 DEG ELBOW LBN WELDED ASTM A1521S16L SCH15S	4.31	1	S98	EL	PR09-0000351
8	90 DEG ELBOW LBN WELDED ASTM A1521S16L SCH15S	4.31	1	S98	EL	PR09-0000351
9	SUP-ON FLANGE FC CL150 ASTM A182 F316S16L	4"	1	S98	FF	PR01-0000260
10	SUP-ON FLANGE FC CL150 ASTM A182 F316S16L	4"	1	S98	FF	PR01-0000260

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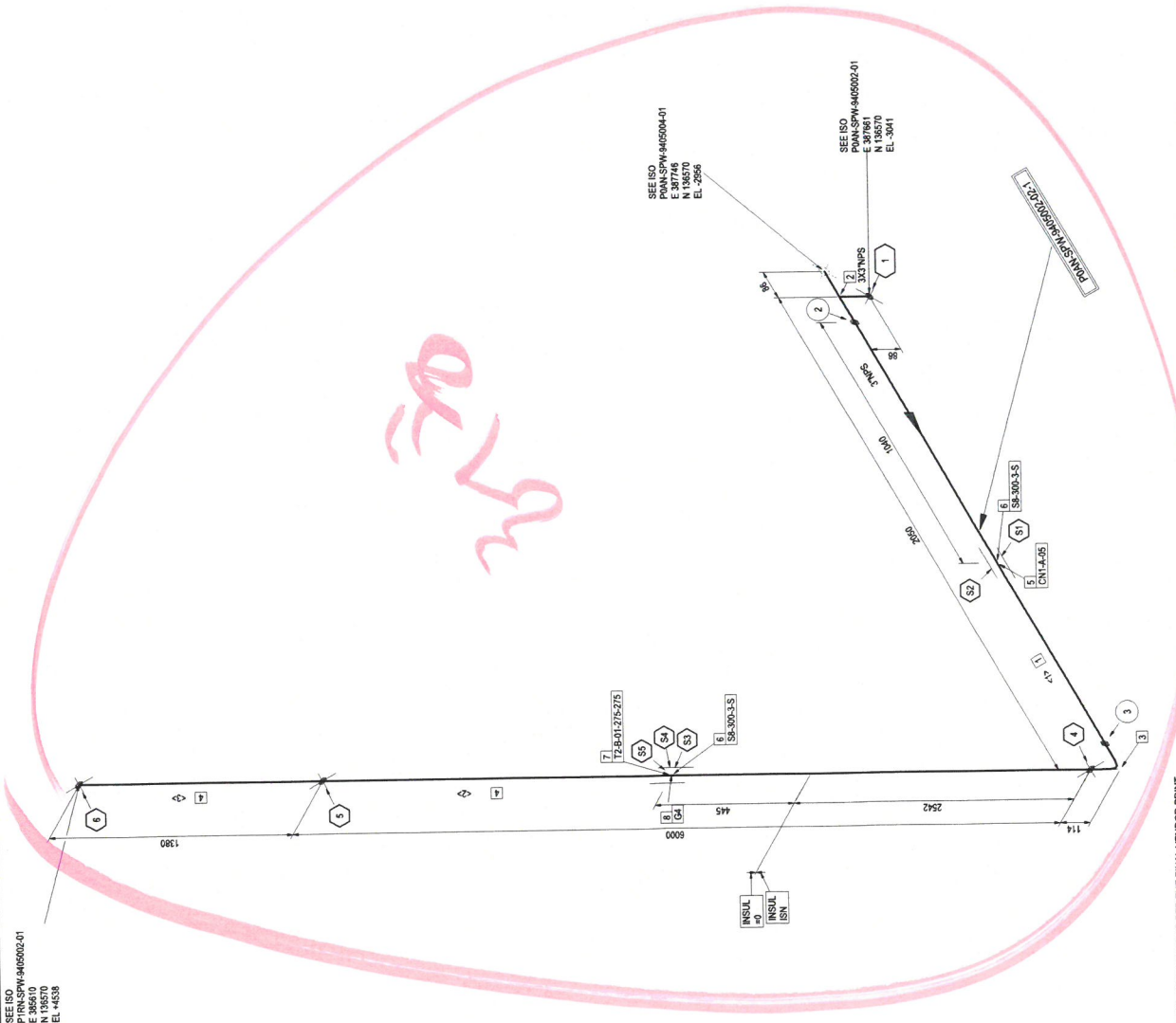
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BILL OF MATERIALS

FABRICATION MATERIALS						
NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE 1/2" SCH 160 SCH 160 BE RT 100%	1/2"	1.9 M	S38	P	PA02-00001903
2	TEE 1/2" WELDED ASTM A403 WP31516L SCH 160	1/2"	1	S38	T	P059-00000681
3	90 DEG ELBOW (L/R) WELDED ASTM A403 WP31516L SCH 160	1/2"	1	S38	E	P059-00000410

ERECTOR MATERIALS						
NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
4	PIPE 1/2" WELDED ASTM A403 WP31516L SCH 160 BE RT 100%	1/2"	7.4 M	S38	P	PA02-00001903
5	CHN A-45	3"	1			CHN-A-45
6	SA-309-3-S	3"	2			SA
7	T2-841-275-275	3"	1			T2-841
8	GA	3"	1			GA

CUT PIPE LENGTH				REMARKS
PRICE	LENGTH	WFO		
NO	MM	NO	(IN)	
1	1600	3	6000	
2	1500	3		



134



Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-04-15

LOTTE BIOLOGICS

LOTTE E&C

PROJECT PROVIDENCE

PART	AG	PLANT	AREA	POB	CWA NO.	POAN
PAID NO	K1-03L-UT-PID-9405		16480			SPW
CAD NO	-		PROJ. NO.	3-SPW-9405002-SPW-1SN	LINE NO.	
DWG NO.	POAN-SPW-9405002-02					REV 0
					SHT NO.	1 OF 1

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BILL OF MATERIALS

FABRICATION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PPE EPW ASTM A312 1/2" SCH 10S BE RT-100%	3"	2.0	S98	P	PR02-0000190
2	PIPE COUPLING SW CL 3000 ASTM A182 F15016	300"	2	S98	N	PR02-0000191
3	PPE NPT F 3/4" SCH 10S BE RT-100%	3/4"	2	S98	NPCS	PR02-0000192
4	CON SWAGE BUSH SMLS ASTM A182 F15016 SCH 10S	3/4"	2	S98	NPCS	PR02-0000193
5	90 DEG ELBOW 1/2" SCH 10S BE RT-100%	1/2"	2	S98	E	PR02-0000194
6	SUP-ON FLANGE 1/2" SCH 10S BE RT-100%	3"	4	S98	F	PR02-0000195

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
7	PPE NPT F 1/2" SCH 10S BE RT-100%	1/2"	1	S98	NPCS	PR02-0000196
8	PIPE COUPLING SW CL 3000 ASTM A182 F15016	300"	1	S98	CPT	PR02-0000197
9	CON SWAGE BUSH SMLS ASTM A182 F15016 SCH 10S	3/4"	4	S98	G	PR02-0000198
10	100 STD BOLT 1/2" X 3/4" SCH 10S BE RT-100%	1/2"	16	S98	B	PR02-0000199
11	CHEK VALVE FLG-45 CL 150 ASTM A182 F15016 SCH 10S	3"	1	S98	VC	PR02-0000200
12	GATE VALVE FLG-45 CL 150 ASTM A182 F15016 SCH 10S	3"	1	S98	VA	PR02-0000201
13	GATE VALVE FLG-45 CL 150 ASTM A182 F15016 SCH 10S	3"	1	S98	VA	PR02-0000202
14	GATE VALVE FLG-45 CL 150 ASTM A182 F15016 SCH 10S	3"	1	S98	VA	PR02-0000203
15	1/2" X 3/4" SCH 10S BE RT-100%	1/2"	1	S98	VA	PR02-0000204
16	1/2" X 3/4" SCH 10S BE RT-100%	1/2"	1	S98	VA	PR02-0000205
17	1/2" X 3/4" SCH 10S BE RT-100%	1/2"	1	S98	VA	PR02-0000206

INSTRUMENTS

18	TAG NO : PG-9405-940-02, INSTRUMENT	3/4"	1	INSTRUMENT
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CUT PIPE LENGTH

NO	LENGTH (MM)	NO	LENGTH (MM)	NO	LENGTH (MM)
1	278	2	278	3	278
4	100	5	100	6	100
7	1500	8	1500	9	1500

REMARKS

NO	REMARKS
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BILL OF MATERIALS

FABRICATION MATERIALS

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PPE EFW ASTM A132-1601HL SCH15S BE RT-100%	3"	1.3M	S98	P	PA02-00001803
2	HALF COUPLING SW CL 3000 ASTM A102-21501HL	3X3/4"	2	S98	TH	PP02-00000851
3	PPE NPTA E PPE SMLS ASTM A102-21501HL SCH15S 10MMX4.7"	3/4"	2	S98	NPTC	PP02-00004539
4	CON SWAGE BLEPSE SMLS ASTM A102-21501HL SCH15S 3X1.12"	3/4"	2	S98	RCS	PP02-00001050
5	90 DEG ELBOW (LR) BW WELDED ASTM A102-21501HL SCH15S	3"	2	S98	E	PP02-00000810
6	SUP CON FLANGE RF CL 150 ASTM A102-21501HL	3"	4	S98	F	PP02-00002656

DIRECTION MATERIALS

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
7	PPE NPTA E PPE SMLS ASTM A102-21501HL SCH15S 10MMX4.7"	3/4"	1	S98	NPTC	PP02-00001050
8	CAP SCRD NPT CL 3000 ASTM A102-21501HL	3/4"	1	S98	CPT	PP02-00000851
9	SPIRAL WOUND GASKET CARBON SS 304 CL 150 PROOF SS 316 FILLER GRAPHITE	3"	4	S98	G	PP02-00000828
10	E GASKET RT HL	3"	1	S98	B	PP02-00001650
11	NS SUD BOLT W/ NUT HEAVY HEX NUT ASTM A193-B8 CL2 ASTM A194-4 UNF	5/8"	16	S98	B	PP02-00001650
12	CHEK VALVE FLG DRF CL 150 ASTM A193-B8 CL2 ASTM A194-4 UNF	3"	1	S98	VC	PP02-00000211
13	GATE VALVE FLG DRF CL 150 ASTM A193-B8 CL2 ASTM A194-4 UNF	3"	1	S98	VA	PP02-00001654
14	OSBY FLEX WDG DISC HW OP (VAN/51)	3/4"	1	S98	VAST	PP02-00001651
15	GATE VALVE SWP SCRD NPT CL 3000 ASTM A102-21501HL SCH15S TRM STELLITE SEATS BB	3/4"	1	S98	VAST	PP02-00001651
16	SEATS BB OSBY SOLID WDG DISC HW OP (VAN/51)	3/4"	1	S98	VAST	PP02-00001651
17	GATE VALVE SW CL 150 ASTM A102-21501HL SCH15S TRM STELLITE SEATS BB OS	3/4"	1	S98	VA	PP02-00001654
18	AT SOLID WDG DISC HW OP (VAN/51)	3"	1	S98	VA	PP02-00001654
19	FLCN AT (INFORMED)	3"	1	S98	VA	PP02-00001654
20	FLCN AT (INFORMED)	3"	1	S98	VA	PP02-00001654
21	BAG-SEWIN W005-246S	3"	1	S98	VA	PP02-00001654

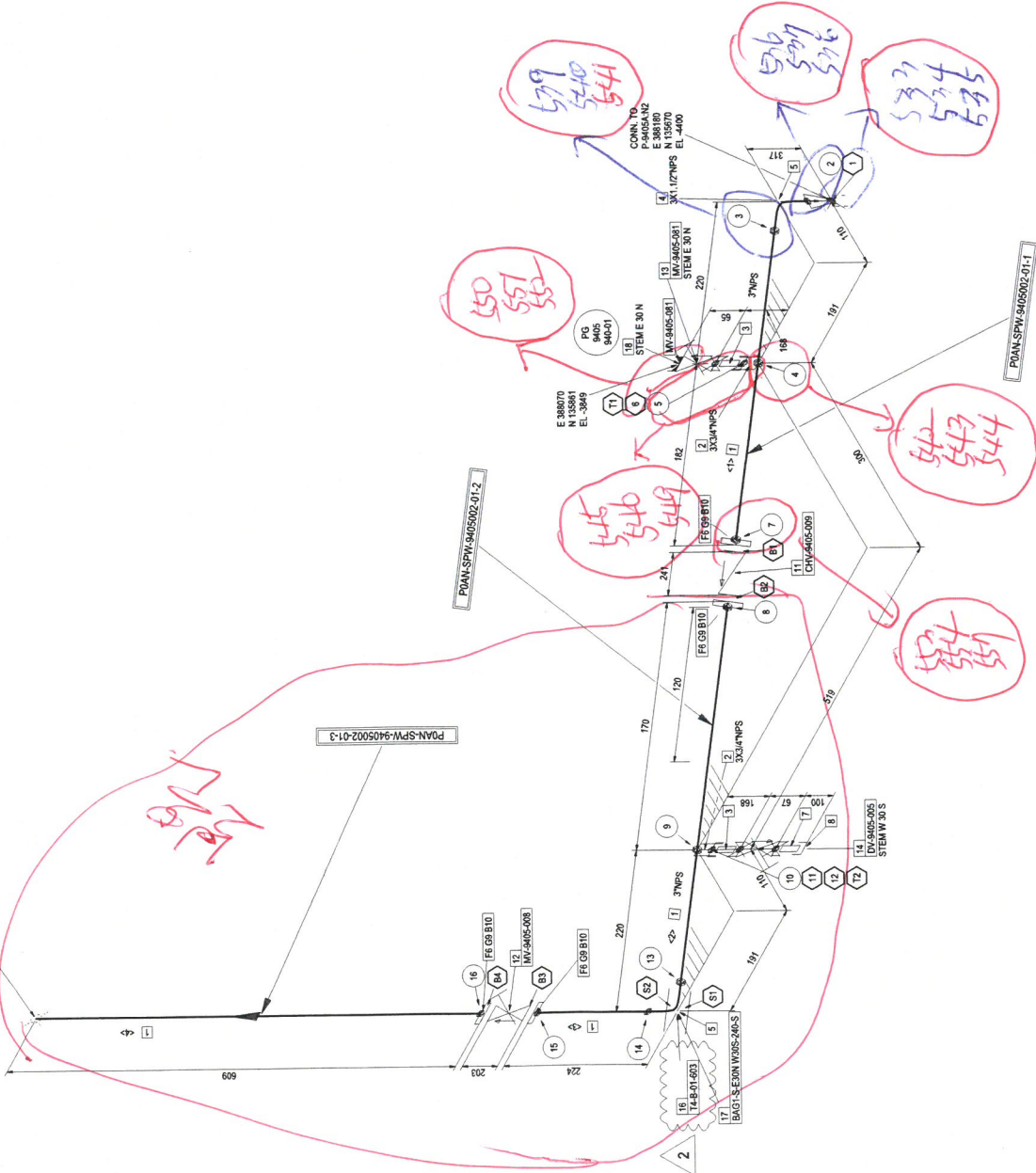
INSTRUMENTS

NO.	INSTRUMENT	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	THIS NO. IS 9405-4401-1 INSTRUMENT	3/4"	1	S98	VA	PP02-00001654

CUT PIPE LENGTH

NO.	LENGTH (MM)	NO.	REMARKS	NO.	LENGTH (MM)	REMARKS
1	278	3	<2>	256	3	
2	100	3	<4>	599	3	

SEE NO.
P0AN-SPW-9405002-02
E 387661
N 138570
EL -3041



NOTE: ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NO.	CLASS	TEMP. (°C)	PRESS. (bar.g)	DESIGN	TEST	INSULATION	PAINT	PHWT	NDE (RT/%)	STRESS	REVISION	DATE	REV.	DESCRIPTION
1	S98	80	5	128	-	S	ISN	25	5	K1-103	1	14-09-25	0	ISSUED FOR REVISION
2	S98	80	5	128	-	S	ISN	25	5	K1-103	2	26-12-25	1	ISSUED FOR REVISION
3	S98	80	5	128	-	S	ISN	25	5	K1-103	3	14-09-25	0	ISSUED FOR REVISION
4	S98	80	5	128	-	S	ISN	25	5	K1-103	4	14-09-25	0	ISSUED FOR REVISION
5	S98	80	5	128	-	S	ISN	25	5	K1-103	5	14-09-25	0	ISSUED FOR REVISION
6	S98	80	5	128	-	S	ISN	25	5	K1-103	6	14-09-25	0	ISSUED FOR REVISION
7	S98	80	5	128	-	S	ISN	25	5	K1-103	7	14-09-25	0	ISSUED FOR REVISION
8	S98	80	5	128	-	S	ISN	25	5	K1-103	8	14-09-25	0	ISSUED FOR REVISION
9	S98	80	5	128	-	S	ISN	25	5	K1-103	9	14-09-25	0	ISSUED FOR REVISION
10	S98	80	5	128	-	S	ISN	25	5	K1-103	10	14-09-25	0	ISSUED FOR REVISION
11	S98	80	5	128	-	S	ISN	25	5	K1-103	11	14-09-25	0	ISSUED FOR REVISION
12	S98	80	5	128	-	S	ISN	25	5	K1-103	12	14-09-25	0	ISSUED FOR REVISION
13	S98	80	5	128	-	S	ISN	25	5	K1-103	13	14-09-25	0	ISSUED FOR REVISION
14	S98	80	5	128	-	S	ISN	25	5	K1-103	14	14-09-25	0	ISSUED FOR REVISION
15	S98	80	5	128	-	S	ISN	25	5	K1-103	15	14-09-25	0	ISSUED FOR REVISION
16	S98	80	5	128	-	S	ISN	25	5	K1-103	16	14-09-25	0	ISSUED FOR REVISION
17	S98	80	5	128	-	S	ISN	25	5	K1-103	17	14-09-25	0	ISSUED FOR REVISION
18	S98	80	5	128	-	S	ISN	25	5	K1-103	18	14-09-25	0	ISSUED FOR REVISION
19	S98	80	5	128	-	S	ISN	25	5	K1-103	19	14-09-25	0	ISSUED FOR REVISION
20	S98	80	5	128	-	S	ISN	25	5	K1-103	20	14-09-25	0	ISSUED FOR REVISION
21	S98	80	5	128	-	S	ISN	25	5	K1-103	21	14-09-25	0	ISSUED FOR REVISION
22	S98	80	5	128	-	S	ISN	25	5	K1-103	22	14-09-25	0	ISSUED FOR REVISION
23	S98	80	5	128	-	S	ISN	25	5	K1-103	23	14-09-25	0	ISSUED FOR REVISION
24	S98	80	5	128	-	S	ISN	25	5	K1-103	24	14-09-25	0	ISSUED FOR REVISION
25	S98	80	5	128	-	S	ISN	25	5	K1-103	25	14-09-25	0	ISSUED FOR REVISION
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27	S98	80	5	128	-	S	ISN	25	5	K1-103	27	14-09-25	0	ISSUED FOR REVISION
28	S98	80	5	128	-	S	ISN	25	5	K1-103	28	14-09-25	0	ISSUED FOR REVISION
29	S98	80	5	128	-	S	ISN	25	5	K1-103	29	14-09-25	0	ISSUED FOR REVISION
30	S98	80	5	128	-	S	ISN	25	5	K1-103	30	14-09-25	0	ISSUED FOR REVISION
31	S98	80	5	128	-	S	ISN	25	5	K1-103	31	14-09-25	0	ISSUED FOR REVISION
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33	S98	80	5	128	-	S	ISN	25	5	K1-103	33	14-09-25	0	ISSUED FOR REVISION
34	S98	80	5	128	-	S	ISN	25	5	K1-103	34	14-09-25	0	ISSUED FOR REVISION
35	S98	80	5	128	-	S	ISN	25	5	K1-103	35	14-09-25	0	ISSUED FOR REVISION
36	S98	80	5	128	-	S	ISN	25	5	K1-103	36	14-09-25	0	ISSUED FOR REVISION
37	S98	80	5	128	-	S	ISN	25	5	K1-103	37	14-09-25	0	ISSUED FOR REVISION
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40	S98	80	5	128	-	S	ISN	25	5	K1-103	40	14-09-25	0	ISSUED FOR REVISION
41	S98	80	5	128	-	S	ISN	25	5	K1-103	41	14-09-25	0	ISSUED FOR REVISION
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77	S98	80	5	128	-	S	ISN	25	5	K1-103	77	14-09-25	0	ISSUED FOR REVISION
78	S98	80	5	128	-	S	ISN	25	5	K1-103	78	14-09-25	0	ISSUED FOR REVISION
79	S98	80	5	128	-	S	ISN	25	5	K1-103	79	14-09-25	0	ISSUED FOR REVISION
80	S98	80	5	128	-	S	ISN	25	5	K1-103	80	14-09-25	0	ISSUED FOR REVISION
81	S98	80	5	128	-	S	ISN	25	5	K1-103	81	14-09-25	0	ISSUED FOR REVISION
82	S98	80	5	128	-	S	ISN	25	5	K1-103	82	14-09-25	0	ISS

BILL OF MATERIALS

FABRICATION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PPE EPW ASTM A153 161.6L SCH10S BE RT=100%	3"	3.5M	S88	P	PA02-00001901
2	45 DEG ELBOW BW WELDED ASTM A403 WP161.6L SCH10S	3"	2	S88	EA	PR06-00003302

ERECTOR MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
3	PPE EPW ASTM A153 161.6L SCH10S BE RT=100%	3"	5	S88	P	PA02-00001901
4	SPL-3	3"	2			SP1
5	G1-02	3"	1			G1
6	DA1	3"	1			DA1
7	SS-300-3-S	3"	1			SS

CUT PIPE LENGTH

NO	LENGTH (MM)	NPD (IN)	REMARKS	PRICE	NO	LENGTH (MM)	NPD (IN)	REMARKS
1	6000	3	<2>	6000	3			
2	1003	3	<4>	625	3			
3			<5>	1819	3			

PROJECT PROVIDENCE

APPROVAL FOR CONSTRUCTION	PIPING MANAGER	Bae Chang Won
2026-01-15		

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LOTTE BIOLOGICS

LOTTE E&C

PROJECT PROVIDENCE

APPROVAL FOR CONSTRUCTION

PIPING MANAGER Bae Chang Won

2026-01-15

SEE ISO YPRN-SPW-9405002-01 E 383725 K 90170 EL +7444

SEE ISO YPRN-SPW-9405002-02 E 383725 K 90170 EL +7444

SEE ISO YPRN-SPW-9405002-03 E 383725 K 90170 EL +7444

SEE ISO YPRN-SPW-9405002-04 E 383725 K 90170 EL +7444

SEE ISO YPRN-SPW-9405002-05 E 383725 K 90170 EL +7444

SEE ISO YPRN-SPW-9405002-06 E 383725 K 90170 EL +7444

SEE ISO YPRN-SPW-9405002-07 E 383725 K 90170 EL +7444

SEE ISO YPRN-SPW-9405002-08 E 383725 K 90170 EL +7444

SEE ISO YPRN-SPW-9405002-09 E 383725 K 90170 EL +7444

SEE ISO YPRN-SPW-9405002-10 E 383725 K 90170 EL +7444

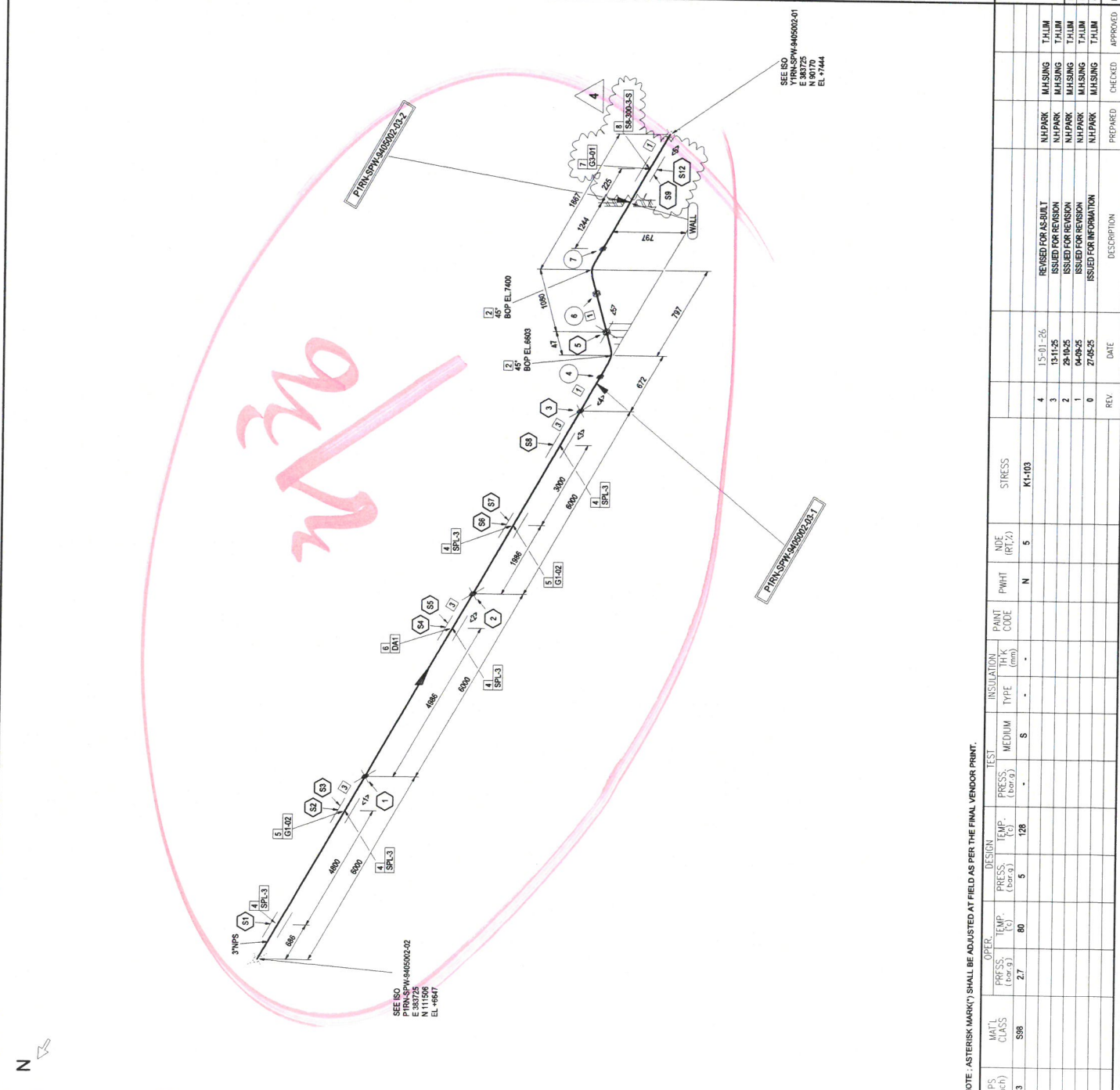
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SEE ISO YPRN-SPW-9405002-14 E 383725 K 90170 EL +7444

SEE ISO YPRN-SPW-9405002-15 E 383725 K 90170 EL +7444



NOTE: (ASTERISK MARK) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

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FABRICATION MATERIALS

13-19

539

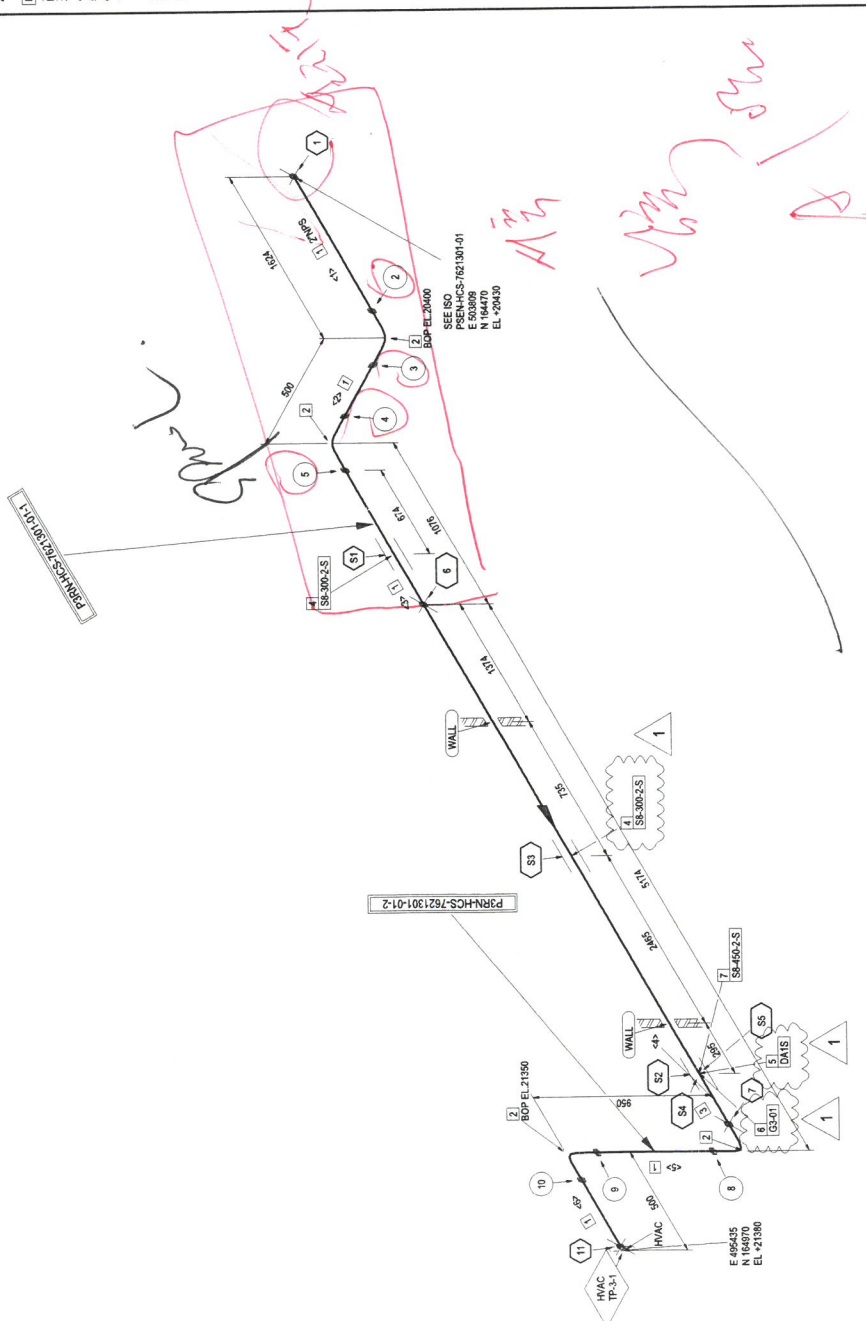


Project PROVIDENCE
Approval For Construction
PIPING MANAGER, Bea Chang Won
2025-09-24



LOTTE E&C

PART	AG	PLANT	N	AREA	PDB	DWA NO.	PRN
PAYD NO	K1-DLSLT-PID-7621-01			16460	FLUID		HCS
CAD NO				PROJ NO.			
DWG NO.				LINE NO.			
				2-HCS-7621-301-S98-AH1			
						SHT NO.	REV
						1	OF 1

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